Food energy is expressed in terms of both kilocalories and kilojoules and represents the physiological energy value remaining after losses in digestion and metabolism have been deducted. (One kilocalorie equals 4.184 kilojoules). A broader discussion on energy and calorie factors used in SR can be found under "Food Description" file in the documentation.

The specific calorie factors used for calculating energy values in pork products are:

	Kcal/g
Protein	4.27
Fat	9.02
Carbohydrate	3.87

The carbohydrate factor of 3.87 is used for estimating energy values for some organ meats and some cured products. The factors are based on the Atwater system for determining energy values. Details of the derivation of these factors are outlined in Agriculture Handbook No. 74 (1973). Because the level of carbohydrate in separable lean and separable fat is insignificant, no carbohydrate factor is needed for these products.

Description of Projects

A series of projects have been conducted to update the pork cuts in the USDA National Nutrient Database for Standard Reference (SR). The studies documented in these notes on pork represent only data collected since 2005. These projects are described in detail below:

Natural Fresh Pork Cuts

Nutrient composition data for fresh pork products in the SR had not been updated since 1991. Since that time, changes in animal husbandry practices and industry procedures led to the availability of leaner cuts. In order to provide up-to-date nutrient information on fresh pork products in SR, the NDL, in collaboration with scientists at the University of Wisconsin and the National Pork Board, conducted a study to determine the nutrient composition of nine (9) fresh pork cuts. This study was funded in part by the National Pork Board. The cuts chosen for evaluation were bone-in shoulder blade steak, boneless tenderloin roast, boneless top loin chop, boneless top loin roast, bone-in sirloin roast, bone-in center loin chop, bone-in center rib chop, bone-in country-style ribs, and bone-in spare ribs. Data from this project were disseminated in a separate report on the NDL web site titled "The Revised USDA Nutrient Data for Fresh Pork" in 2006 and were later incorporated in SR20 (2007). New data obtained to update existing data for pork top loin chops, raw and cooked (broiled), are disseminated in SR 27(2014).

Sampling: Nine fresh pork cuts were pre-ordered and purchased from 12 retail outlets using the nationwide sampling plan developed for NFNAP (Perry *et al*., 2003) and

shipped frozen to the University of Wisconsin for trimming and preparation. Products from each location were assigned randomly to either raw or cooked preparation. For roasts and spare ribs, each roast or rack of ribs was randomly assigned to either raw or cooked preparation.

Preparation - Cooking procedures:

Broiling (Center Loin Chops, Center Rib Chops, Top Loin Chops). Chops were grilled on a pre-heated George Foreman™ Indoor/Outdoor Electric Barbeque Grill for 10 minutes, setting "4". External fat thickness and chop thickness were measured prior to cooking; weights of raw cuts were obtained. Two (2) thermocouples were placed into one (1) or two (2) chops, as needed. Chops were turned over when the internal temperature reached 100°-105°F (38°-41°C). Chops were removed from the grill to attain a final internal temperature of 160°F/71°C (chops were taken off the grill at approximately 155°F/68°C internal temperature). Chops were cooled on a wire rack for 5 minutes and the highest internal temperature attained during the standing period was recorded. After standing for 5 minutes, chops were re-weighed.

Roasting (Top Loin, Tenderloin, and Sirloin Roasts). Oven was pre-heated to 325°F/163°C (425°F/218°C for tenderloin roast). Top loin, tenderloin, and sirloin roasts were weighed raw, and placed on a rack in a pan for cooking. Top loin roasts (boneless) were roasted as "single" loin roasts (one loin muscle only). If the purchased product was "double top loin roast (boneless)", i.e. two single top loin roasts backed and tied together, the strings were removed, and each half of the double top loin roast was processed as a single top loin roast. Roasts were cooked uncovered. An oven-durable meat thermometer was placed into the geometric center of the roast. Roasts were removed when they achieved an internal temperature of ~150°F/65°C; the target final internal temperature was approximately 160°F/71°C. Roasts were allowed to stand 15 minutes; the final internal temperature was determined during this period. The cooked weight of the roast was obtained and the cooking yield calculated.

Roasting (Spareribs). The oven was pre-heated to 325°F/163°C. No external fat measurements were collected, but any gross physical fat (loosely attached) from the raw ribs were removed before cooking. The raw weight of the spareribs was obtained. The number of ribs in the product being cooked was recorded. Spareribs were placed on a rack in a pan, but were not covered during cooking. Ribs were roasted for 1 hour and 45 minutes. Ribs were then removed from the oven; the temperature in the intercostal muscles was immediately taken. Ribs were cooled for 10 minutes, and then re-weighed. When cool enough to process, edible lean was separated from bone/cartilage. Trimmable fat and connective tissue are not an issue in cooked ribs, since it is assumed that, with this product, all soft tissues are consumed.

Braising (Shoulder Blade Steaks and Country-Style Ribs). Oven was pre-heated to 325°F/163°C. The raw blade steaks and/or country-style ribs were weighed. The thickness of the external fat around the outer surface of the cuts was measured. Blade

steaks or country-style ribs were placed on a rack in a roasting pan. Distilled water (100 ml) was added to the roasting pan, which was covered tightly and placed in the center of the oven. Cooking time was determined from initial trials. Initial cooking time estimates were: 45 minutes for blade steaks; 1 hour and 15 minutes for country-style ribs. The internal temperature was determined with an electronic digital thermometer. Steaks and/or ribs were allowed to cool for 5 minutes and then re-weighed and the weight was recorded.

Sample preparation - raw and cooked products:

Measurement of external trim (separable) fat. For all chops, blade steaks, and country-style ribs, external fat at the 1/4", 1/2" and 3/4" points along the external fat surface of the product were measured in millimeters. External fat thickness was measured at each of these points. For top loin and sirloin roasts, fat thickness measurements were taken over the center of the exposed fat at the 1/4", 1/2" and 3/4" points along the length of the roast. External fat measurements were not determined on tenderloin roast or spareribs.

Separation of lean meat, separable fat, connective tissue, and bone. Dissection of pork cuts was performed from the perspective of a "careful consumer", who conscientiously separates these tissues. The most difficult separation is between the trimmable (separable) fat and connective tissue, which lies in the "seams" between muscles. The separation was accomplished by "scraping" the co-mingled tissues with a knife blade, such that the soft fat was separated from the tougher, stringy connective tissue. Separable lean tissue should be relatively free of trimmable fat, while the trimmable fat should be reasonably free of connective tissue.

Separable lean meat, separable fat, and connective tissue were removed from bones as cleanly as possible. Separable fat (i.e., external trim fat and seam fat), bone, and connective tissue were removed from raw and cooked products and weighed to determine the relative amounts of separable fat and separable lean meat. Component weights (i.e., weights of separable lean, separable fat, bone, and connective tissue) were reported in SR; weights of connective tissue and bone were combined and reported as "refuse". For food items listed "lean only", the separable fat associated with that cut is considered "refuse"; for food items listed "lean and fat", the separable fat is considered edible and contributes to the nutrient profile.

Sample composites and nutrient analyses:

Shoulder blade steak, tenderloin roast, and top loin chops. Shoulder blade steak, tenderloin roast, and top loin chops represent different areas of the pig and are most commonly cooked by grilling, roasting, and braising, respectively. For purposes of this study, these were referred to as the primary cuts since complete nutrient profiles were obtained for both the raw and cooked preparations of these cuts. For each cut, the lean tissue cuts purchased from an individual location were combined into individual composites for homogenization and nutrient analysis; for some nutrients (proximates,

minerals, cholesterol, thiamin, niacin, and riboflavin), the number of observations (n) = 12. For pantothenic acid, vitamin B_6, and vitamin B_{12}, samples from the three locations were combined to form regional composites (n = 4). One of these composites was randomly chosen and analyzed for retinol (Vitamin A); n = 1. Separable fat from all cuts were combined to form raw and cooked composites. Complete nutrient profiles were determined for each of these composites (raw and cooked).

Top loin roasts, sirloin roasts, center loin chops, center rib chops, country-style ribs, and spare ribs. Proximate nutrients and minerals were analyzed from individual composites for both the raw and cooked preparations of top loin roasts, sirloin roasts, center loin chops, center rib chops, country-style ribs, and spare ribs. For these cuts, cholesterol, thiamin, niacin, and riboflavin were determined from the regional composites of the cooked samples. For some nutrients, values were imputed using established NDL procedures described above. Nutrient values for pantothenic acid, vitamin B_6, and vitamin B_{12} for these cooked cuts were imputed from the primary cuts prepared (cooked) in the same manner. Nutrient values (cholesterol, thiamin, niacin, and riboflavin, pantothenic acid, vitamin B_6 and vitamin B_{12}) for the raw preparations were imputed from their cooked counterparts. A commercial laboratory, whose analytical procedures were evaluated through the NFNAP process and found to be acceptable, performed tissue homogenization and nutrient analyses.

Enhanced Pork Cuts

Enhanced pork is the process of adding non-meat ingredients to fresh pork to improve the eating quality of the final product where eating quality is defined as the juiciness, tenderness, and flavor of pork (National Pork Board, 1998). As meat producers increasingly raise leaner animals that contain significantly less fat, alternative processes are being developed to replace the flavor loss due to fat reduction and reduce moisture loss resulting from cooking. Enhancing the meat is one such process. Since SR did not provide data for the nutrient content of enhanced meat, a collaborative study was conducted by scientists at USDA, the University of Wisconsin, and the National Pork Board to determine the nutrient profile of the following enhanced products: shoulder blade steak, tenderloin, and top loin chops. This project was funded in part by the National Pork Board. Data for enhanced pork cuts were disseminated in SR20 (2007). New data obtained to update existing data for enhanced pork top loin chops, raw and cooked (broiled) are disseminated in SR 27 (2014).

Sampling. Three fresh, enhanced pork cuts were pre-ordered and purchased from 12 retail outlets using the nationwide sampling plan developed for NFNAP (Perry *et al*., 2003) and shipped frozen to the University of Wisconsin for trimming and preparation.

Preparation and analysis. Preparation, compositing, and nutrient analyses for enhanced versions of the shoulder blade steak, tenderloin, and top loin chops were similar to those described for natural fresh pork cuts (see above).

Pork Value Cuts

USDA, in collaboration with the National Pork Board and University of Wisconsin, conducted a study to determine the nutrient profile of four new pork value cuts. This project was funded in part by the National Pork Board. These cuts were introduced to the retail market in 2008-2009. Pork value cuts are individual muscles chosen from the shoulder and the leg. These cuts were selected for their strong marketability, consistency in flavor and tenderness, availability, and economic feasibility for food chains and consumers. The common names of the four new cuts selected, the scientific name for the muscle, and the part of the carcass from which they originate are as follows:

- Pork Shoulder Breast Boneless (Pectoralis profundi) – shoulder
- Pork Shoulder Petite Tender Boneless (Teres major) - shoulder
- Pork Leg Cap Steak Boneless (Gracilis) – leg
- Pork Leg Sirloin Tip Roast Boneless (Vastus lateralis and Rectus femoris) – knuckle and leg.

The nutrient profiles of these four new cuts were released in SR21 (2008).

Sampling. A total of 14 paired cuts for each pork value cut were obtained from pork production plants in North Carolina and Iowa. At each plant, both shoulder and hams from 7 randomly selected pork carcasses were obtained. Carcasses were of average weight or slightly heavier to ensure an adequate amount of sample. Proper cut identification of each ham and shoulder from each plant was maintained throughout the fabrication process. Each muscle was denuded, trimmed free of all external fat and connective tissue, and frozen prior to shipment to the University of Wisconsin.

Sample Preparation. Among the 7 paired products from each of the two locations, 6 pairs were randomly selected for use in the study. One member of each pair was prepared as raw and the other was cooked either by broiling or braising to a desired internal temperature or time end-point. After a designated cooling period, the cooked product was cubed, hand mixed, and divided into individual carcass samples, and composites of two or three carcasses.

The designated cooking method for each pork value cut were:

- Pectoralis profundi – broiled
- Teres major – broiled
- Gracilis – broiled
- Rectus femoris – braised

Cooking methods, broiling. Cuts were grilled on a pre-heated George Foreman™ Indoor/Outdoor Electric Barbeque Grill for 10 minutes on setting “4”. Raw cuts were weighed prior to cooking. Internal cooking temperatures were determined by insertion of thermocouples. Cuts were turned-over when the internal temperature reached 100°-105°F (71°-41°C). Cuts were removed from the grill to attain a final internal temperature of 160°F/71°C (cuts were taken off the grill at approximately 155°F/68°C internal

temperature). After standing 5 minutes, cuts were re-weighed and the highest internal temperature was attained during the standing period and recorded.

Cooking methods, braising. Oven was pre-heated to 325°F/163°C. Temperature was monitored with an oven thermometer. The cuts were weighed prior to cooking and then placed on a rack in a roasting pan. Distilled water (100 ml) was added to the roasting pan, which was covered tightly and placed in the center of the oven. Cuts were braised until reasonably tender. Cooking time was determined from initial trials. Initial cooking time estimates were: 45 minutes for blade steaks; 1 hour and 15 minutes for country-style ribs. Immediately after removal from the oven, the product was placed on a wire rack. The internal temperature was determined with an electronic digital thermometer. Cuts were allowed to cool for 5 minutes and then weighed.

Sample analyses. Proximate nutrients (moisture, total fat, ash, and protein) and cholesterol were determined on individual muscle samples from the shoulder, leg and knuckle, both raw and cooked. For each cut, three samples were pooled into composites and analyzed for fatty acids. Vitamins and minerals were analyzed on samples from the two-carcass composites. Choline and folate analyses were done on the three-carcass composites, raw and cooked. Amino acids were also analyzed on the three-sample composites - raw samples only.

Cured Hams

A new study on cured ham products was conducted by the NDL in collaboration with the University of Wisconsin to update the nutrient profile of various cured ham products in the SR. The word Ham refers to pork meat from the hind leg of a hog. Ham products were available in bone-in or boneless forms.

Cured hams are classified into four categories (USDA-FSIS, 2007):

- Ham - at least 20.5% protein in the lean area with no water added;
- Ham with Natural Juices (HNJ) - at least 18.5% protein with a small addition of water when cured;
- Ham - Water added (HWA) - at least 17% protein with no more than 10% added solution;
- Ham and Water Product (HWP) - less than 17% protein and contains any amount of water but labeling must indicate percentage of “added ingredients”.

“Added ingredients” may vary for each ham product. These solutions, flavorings or “added ingredients” may include water, sugar, salt, sodium erythrobate, sodium nitrite, potassium, and magnesium leading to flavor enhancement. Binders such as soy or milk proteins may also be added to help hold water in the ham. These additions of water and flavor enhancers in ham affect its taste and texture.

Sampling. The sampling plan used for the study was developed for NFNAP (Pehrsson *et al*., 2000). The country was divided into four regions, with three consolidated

metropolitan statistical areas (CMSA) within each region; two retail stores were selected within each CMSA. Eight different types of ham products were picked up from 12 retail outlets nationwide: 1) ham, bone-in whole; 2) ham, bone-in, shank half; 3) ham with natural juices, bone-in rump; 4) ham with natural juices, bone-in butt half; 5) ham with natural juices, bone-in spiral sliced; 6) ham, water added, bone-in, slice; 7) boneless hams (many shapes and sizes); and 8) ham and water product, boneless slices, any type, and/or glazed with sugar, honey, and other ingredients. The sampling procedure for each category of bone-in hams was to select two half-hams. One of those was a shank-half portion and the other a rump-half portion. It was preferable that the two halves should come from the same manufacturer and from the same category. Pairs of selected, branded, bone-in hams (Maple, Haen, and Brandon) were picked-up for retention studies. All products were vacuum packaged, individually labeled, and sent frozen to University of Wisconsin for further cooking and dissection.

Sample preparation. All hams (bone-in and boneless; heated and unheated) were weighed, measured for thickness, and dissected to separate external fat and seam fat. Bone-in hams were further dissected for removal of bone and connective tissue prior to nutrient analyses. Branded hams or paired bone-in whole hams were cut into shank, butt, and slices. One portion from each pair (rumps and shanks) was analyzed "as purchased" and the other roasted to an internal temperature >160°F (71°C). Slices were weighed and measured for thickness prior to being pan-fried to an internal temperature of 64-82°F (18°-28°C). All other types of bone-in and boneless hams were either roasted in a 325°F (163°C) convection oven or pan-broiled to the internal temperature specified on the label. No fat was added during any cooking preparation.

Sample analyses. Proximate nutrients (moisture, total fat, ash, and protein) cholesterol, vitamins, and minerals were determined on all categories of bone-in and boneless hams, both heated and unheated. Total sugars and fatty acids were analyzed on all bone-in and boneless forms of "Ham", "Ham with natural juices" and "Ham and water product". Two pairs of "Ham" types, heated and unheated, were analyzed for vitamin K, retinol, choline, and amino acids (unheated only).

References for Notes on Foods – Pork Products

Jones DB. Factors for converting percentages of nitrogen in foods and feeds into percentages of protein. US Department of Agriculture, Circular 83, slight revision, 1941.

Merrill AL, Watt BK. Energy value of foods: basis and derivation, revised. US Department of Agriculture, Agriculture Handbook 74, 1973.

National Pork Board. Functionality of non-meat ingredients used in enhanced pork. Version current 1998. Internet: http://www.pork.org/filelibrary/Factsheets/PorkScience/functionalitynonmeat04527.pdf (accessed 30 June 2014).

Perry CR, Pehrsson PR, Holden J. A revised sampling plan for obtaining food products for nutrient analysis for the USDA national nutrient database 2003. Proceedings of the American Statistical Association, Section on Survey Research Methods [CD-ROM]. 2003. Alexandria, VA: American Statistical Association, San Francisco.

Pehrsson PR, Haytowitz DB, Holden JM, Perry CR, Beckler DG. USDA's national food and nutrient analysis program: food sampling. J Food Comp Anal 2000;13:379-389.

US Department of Agriculture, Food Safety and Inspection Service. Ham glossary – Ham and food safety fact sheet. Version current 2013. Internet: http://www.fsis.usda.gov/wps/portal/fsis/topics/food-safety-education/get-answers/food-safety-fact-sheets/meat-preparation/ham-and-food-safety/ct_index (accessed 30 June 2014).

Poultry Products (Food Group 05)

Introduction

Data for poultry products including chicken, turkey, and other fowl are presented in the USDA National Nutrient Database for Standard Reference (SR). For most listings of fresh chicken or turkey, nutrient values are given either for specific market types such as broiler or fryer chicken, young tom turkey, young hen turkey, or as "all classes" to indicate that the values represent one or more unidentified market types for the species.

The data in SR represent the amount of each constituent in 100 grams of edible portion. Edible portion for poultry may be represented as "meat and skin" or as "meat only". In both cases, bone, cartilage, and separable fat are removed from the meat and reported as refuse. Weights are determined for the whole bird as purchased and as parts (breast, thigh, wing, drumstick, back), skin, and refuse (bone and cartilage). Refuse is reported as percentages in the SR Full Report format.

Nutrient analyses are conducted for meat and for skin in both raw and cooked forms. Nutrient values for meat only and for skin only are weighted proportionally, according to their respective contributions to each intact bird part (as purchased), or part of the whole bird, and then reported as "meat and skin." Cooked poultry has been cooked with the skin intact, unless otherwise indicated.

The analytical nutrient data include the mean nutrient value, the standard error given to three decimal places, and the number of observations on which the values are based. For many food items, mean values are given without an accompanying standard error and number of samples. These values are either calculated by pooling data by or by weighting means, by applying cooking yields and/or nutrient retention factors, or by imputation from a different, closely related food. For poultry, nutrient estimates are based on the nutrient value for lipids (e.g., fatty acids), total solids (e.g., cholesterol), moisture-free, fat-free solids (e.g., minerals), or protein (e.g., water-soluble vitamins) fraction of a similar food.

Nutrients

Nutrient information for SR can be found under "File Content" in the documentation. In addition, nutrient information which is specific to poultry products is provided in this report. Nutrient data are obtained for moisture, protein, ash and total fat. The values for protein are calculated from the content of total nitrogen (N) in the food using the conversion factor recommended by Jones (Jones, 1941). The specific factor for protein applied to poultry items is 6.25. The carbohydrate content of uncured products consisting entirely of poultry (except some organ meats) is negligible. For such foods, the carbohydrate content is assumed to be zero.

For heart, liver, giblets, and cured poultry products (foods which are expected to contain carbohydrate), the carbohydrate value is calculated as the difference between 100 and the sum of the percentages of water, protein, total lipid, and ash. If the total of these constituents for an item is more than 100 due to analytical variation, the carbohydrate content is assigned a zero value.

Food energy is expressed in terms of both kilocalories and kilojoules; one kilocalorie equals 4.184 kilojoules. The data are the physiologic energy values which remain after losses due to digestion and metabolism have been deducted. Further discussions on energy and caloric factors used in SR can be found in the "Food Description File" of the general documentation.

The specific calorie factors used for calculating energy values in poultry products are:

	Kcal/g
Protein	4.27
Fat	9.02
Carbohydrate	3.87

The carbohydrate factor of 3.87 is used for some organ meats and some cured products. Because the level of carbohydrate in poultry meat and skin is insignificant, no carbohydrate factor is needed for most poultry products. The factors are based on the Atwater system for determining energy values. Details of the derivation of these factors are outlined in Agriculture Handbook No. 74 (Merrill, 1973).

Description of Projects

The studies documented in these poultry notes represent data collected since 2008.

Rotisserie chicken

In collaboration with Texas Tech University, the USDA conducted studies to determine the nutrient composition of commercially prepared rotisserie chicken. The studies were conducted because rotisserie chicken has become a popular ready-to-serve poultry item in the retail market. A study was conducted to obtain data for rotisserie chicken original flavor, released in SR21 (2008) and rotisserie BBQ flavor chicken released in SR25 (2012). New data on rotisserie chicken original flavor were obtained to update the existing values and were released in SR27 (2014).

Sampling: Rotisserie chicken samples were procured nationwide from 12 retail locations using the National Food and Nutrient Analysis Program (NFNAP) nationwide sampling plan developed for the USDA (Perry et al, 2003).

Sample preparation: Samples were purchased whole and dissected into separate parts: breast, thigh, wing, drumstick, and back. Each bird was weighed whole. After dissection, parts were weighed separately, with and without skin. Refuse including